CAPECITABINE

DPD testing
Hand–foot syndrome

CHEMICAL NAME Pentyl [1-(3,4-dihydroxy-5-methyltetrahydrofuran-2-yl)-5-fluoro-2-oxo-1H-pyrimidin-4-yl]carbamate
CLASS Oral fluoropyrimidine (prodrug of 5-FU)
TRADE NAMES Xeloda · Capecitabine (generic) · Capegard

THERAPEUTIC USES / INDICATIONS

- Colorectal cancer (adjuvant and metastatic, CAPOX).
- Breast cancer (metastatic).
- Gastric, oesophageal, pancreatic cancers; rectal cancer with radiotherapy.

ADMINISTRATION

- **PO** twice daily within 30 min after a meal, with water; swallow whole.
- Typically 14 days on, 7 days off (21-day cycle) or continuous with radiotherapy.
- Cytotoxic — handle with gloves; do not crush.

DOSAGES

- **Monotherapy:** 1,250 mg/m² twice daily for 14 days, every 21 days.
- **CAPOX:** 1,000 mg/m² twice daily days 1–14.
- **With radiotherapy:** 825 mg/m² twice daily continuously.
- **CrCl 30–50:** 75% dose; <30: contraindicated.

PHARMACOKINETICS

- Rapidly absorbed; converted in liver and tumour tissue (thymidine phosphorylase) to 5-FU.
- Excreted in urine. **t½:** 0.75 h.

PHARMACODYNAMICS

- Delivers **5-FU** preferentially to tumour → thymidylate synthase inhibition.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hand–foot syndrome.
- Diarrhoea, nausea, stomatitis.
- Fatigue, anorexia.

ADVERSE EFFECTS (SERIOUS)

- DPD-deficiency severe toxicity.
- Severe diarrhoea/dehydration → AKI.
- Cardiotoxicity (angina, MI).
- Myelosuppression (less than 5-FU bolus); hyperbilirubinaemia.

CONTRAINDICATIONS & CAUTIONS

- Complete DPD deficiency; brivudine within 4 weeks.
- Severe renal impairment (CrCl <30).
- Pregnancy, breastfeeding.

KEY INTERACTIONS

- Warfarin: ↑↑ INR, bleeding; monitor closely or switch to LMWH.
- **Phenytoin:** ↑ levels.
- **Brivudine:** contraindicated.

NURSING MANAGEMENT

- Verify DPD testing.
- Teach dosing schedule and toxicity grading; when to stop tablets.
- Assess hands/feet, stools, mouth each visit.
- INR monitoring if on warfarin.
- FBC, renal function, bilirubin before cycles.

PATIENT EDUCATION

- Take within 30 minutes after breakfast and dinner.
- Stop the tablets and phone the team for 4+ extra stools/day, painful red/peeling hands/feet, mouth sores preventing eating, vomiting, fever.
- Moisturise hands/feet daily; avoid heat and friction.
- Do not double missed doses.

KEY POINTS

- Oral 5-FU prodrug (tumour-activated).
- Hand–foot syndrome, diarrhoea.
- 1,250 mg/m² BD × 14 days per 21.
- DPD testing; raises INR with warfarin.